



Auditing AI-Generated Draft Judgments

Citation Provenance, Holding Accuracy, and Reasoning Grounding

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Background: why draft judgments need auditing

The problem

- Large language models are increasingly proposed for judicial decision support, and regulators now treat judicial AI as high-risk.
- Such systems can fabricate authorities, misstate what real cases held, or produce reasoning grounded in no retrieved source.
- In a judicial context these failures bear directly on fairness and the rule of law.

What prior work covers

Magesh et al. (2024) [1] commercial legal-research tools produce fabricated or unsupported citations at measurable rates, but US case law, short-form QA.

Dahl et al. (2024) [2] leading LLMs produce legal hallucinations at rates exceeding 50%, even on specific, verifiable questions about real cases.

Gao (2024) [3], Es (2024) [4], Rashkin (2023) [5] RAG evaluation separates retrieval quality from generation faithfulness, and asks whether each statement is attributable to identified sources.

UK Judicial Office (Oct. 2025) [6] permits AI for summarisation but warns that AI can generate fictitious cases and quotes, and that all output must be independently verified.

The gap: no structured, citation-level audit of a UK legal-AI system producing full draft judgments.

Research question and objectives

When Hercules generates a draft judgment, can the cited authorities be traced, are they described accurately, and is the reasoning supported by retrieved evidence?

These are distinct failure modes: a citation can be traceable yet misdescribed, and a correctly described authority can still be attached to reasoning the evidence does not support.

Objectives

1. Extract and classify citations from the AI-generated draft judgments.
2. Trace each citation to the frozen corpus registry and linked retrieval records, and verify existence against BAILII [7].
3. Assess holding accuracy through structured annotation with adjudication.
4. Assess reasoning grounding against the passages actually retrieved.
5. Build and validate an automated screening verifier.

Scope: *the controlled retrieval-depth, prompt-design and model-comparison experiments set out in the original plan were not conducted, and are carried forward as future work.*

The Hercules system and the audited corpus

A proof-of-concept legal-AI system from Ulster University's Centre for Legal Technology, funded by the UK AI Security Institute. It is a research prototype and is not deployed in any court.



Structured data: a separate Neon/PostgreSQL registry of 2,541 case records used for citation matching. Qdrant is the retrieval knowledge base and Neon the structured record set, so their coverage differs by design.

The audited corpus

59

draft judgments
34 real · 25 synthetic

115

citation occurrences
110 case-law · 5 statutory

35

distinct authority
groups

98

validation occurrences
from 50 judgments

Terms: a holding is the legal principle a case establishes, as distinct from its facts or outcome. BAILII [7] is the free public repository of UK and Irish judgments.

A three-layer audit framework

Layer 1 — Citation provenance

Can each citation be traced to the frozen Neon registry and to the retrieval linked to that generation?

Deterministic lookup against frozen audit records. Six statuses, from retrieved-and-cited to not-found-in-registry.

Layer 2 — Holding accuracy

When the system cites a real case, is the legal principle it attributes to that case accurate?

Every distinct authority checked against BAILII [7] and official sources. Generated principle compared with the relevant paragraphs of the full judgment.

Layer 3 — Reasoning grounding

Is the proposition applied in the generated text supported by a passage the system actually retrieved?

A yes label required a retrieved passage to contain or directly entail the premise. Indirect topical relevance was not sufficient.

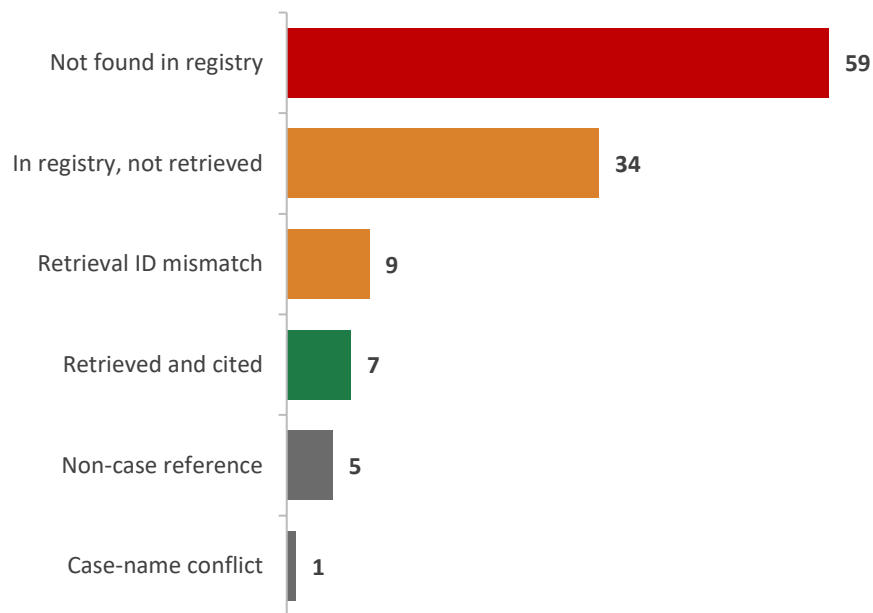
Annotation: primary human pass (R1), AI-assisted second pass (R2), author adjudication. Labels: yes / partial / no / unclear / n-a.

Validation sample: 50 judgments, 98 citation occurrences.

The three layers implement a domain-specific version of attribution to identified sources: Layer 1 checks source identification, Layer 2 claim accuracy, Layer 3 evidential support.

Layer 1: most citations lack a provenance trail

59 of 110 case-law citations (53.6%) were not found in the frozen registry and were not present in the linked retrieval.



Counts across all 115 citation occurrences. Case-law percentages in the text use $n = 110$, excluding the five non-case references.

No fabrication confirmed

Of 33 distinct case-law authorities, 32 were located in external sources and one remained ambiguous. None was confirmed as fabricated.

Citation form (35 groups)

23 correct and complete (65.7%)

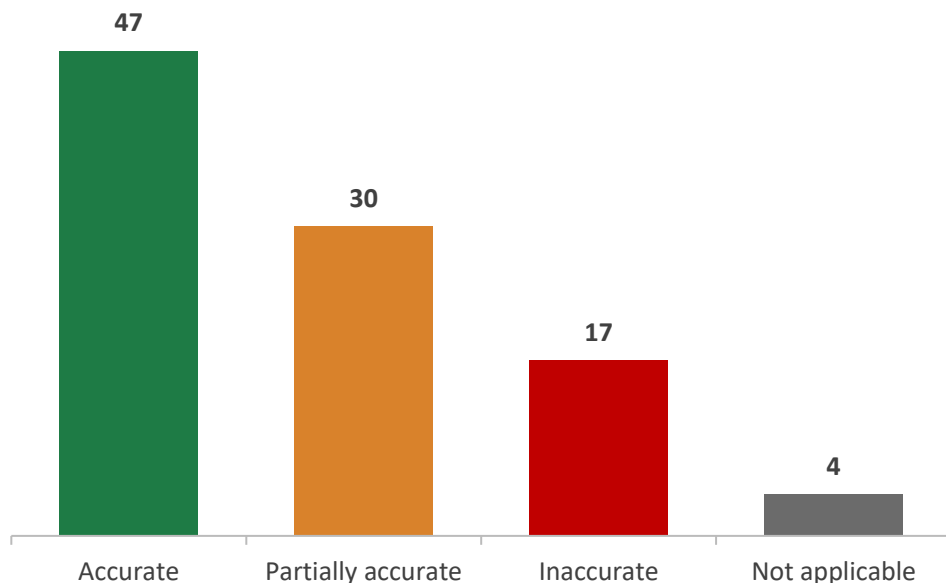
8 incomplete (22.9%)

4 incorrect (11.4%)

This is a provenance gap, not proof of absence: because Qdrant and Neon have different coverage, it does not establish that all 59 authorities were missing from the knowledge base.

Layer 2: half of holding attributions were defective

47 of 94 assessable attributions (50.0%) were inaccurate or only partially accurate.



Final adjudicated labels across all 98 validation occurrences.

Excluding the four n-a cases

47 accurate (50.0%)

30 partially accurate (31.9%)

17 inaccurate (18.1%)

Defect rate 50.0%, 95% CI 40.1–59.9.

At judgment level

29 of 50 validation judgments (58.0%) contained at least one holding defect.

Errors ranged from imprecise descriptions of a relevant principle to authorities drawn from wholly unrelated areas of law.

What a holding error actually looks like

Gorringe v Calderdale Metropolitan Borough Council [2004] UKHL 15 : retrieved and cited

Hercules: a highway authority has a duty to ensure road safety.

The authority: the House of Lords held the council did not owe the claimant a duty to place a road marking or warning sign at the accident location.

The effect of the decision is reversed. Cited twice; both attributions were labelled inaccurate.

Chartbrook Ltd v Persimmon Homes Ltd [2009] UKHL 38 : retrieved and cited

Hercules: courts may imply reasonable pricing terms where the contract is unclear.

The authority: a decision concerning contractual interpretation and the correction of drafting language, rather than the implication of reasonable pricing terms.

Wrong principle. Also labelled ungrounded in the reasoning annotation.

Chapman v Mid & South Essex NHS Foundation Trust [2023] EWHC 1290 (KB) : not in registry, not retrieved

Hercules: statutory control over public safety for cyclists and pedestrians.

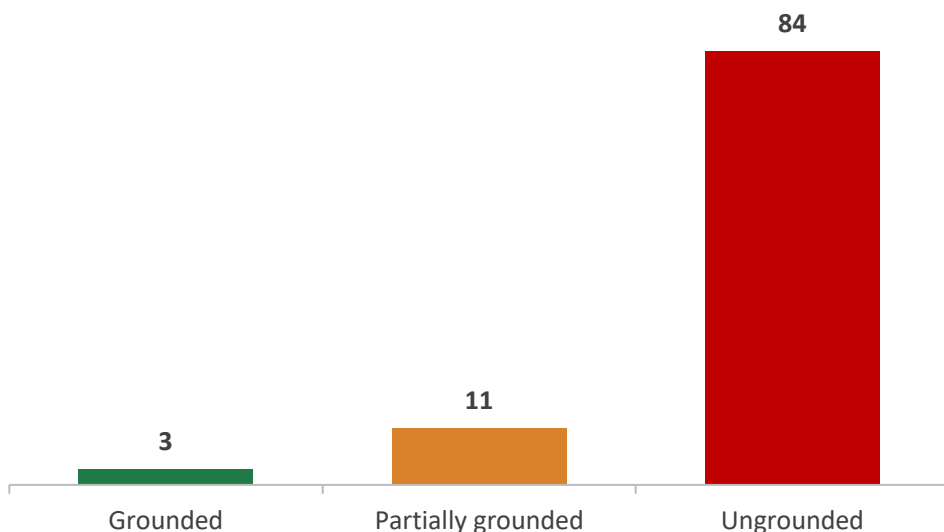
The authority: a clinical-negligence case concerning delayed diagnosis and treatment of a spinal condition, unrelated to highway safety.

Wrong area of law.

**Gorringe and Chartbrook were both retrieved and cited, yet both were misdescribed.
The failure cannot be explained by retrieval absence alone.**

Layer 3: reasoning not grounded in retrieval

84 of 98 reasoning occurrences (85.7%) were not supported by any passage the system actually retrieved.



Final adjudicated labels, 98 validation occurrences. 95% CI 77.4–91.3.

At judgment level

48 of 50 validation judgments (96.0%) contained at least one ungrounded reasoning step.

What this means

The system produces confident reasoning that reads like conventional judicial writing.

But it is constructing plausible-sounding reasoning rather than reasoning from its sources.

Building the automated verifier

A Python verifier reads the annotation records, the linked retrieval evidence and the frozen Neon registry, and produces a separate screening prediction for each of the three layers.

Layer 1 : citation matching

Matches neutral and law-report citations against the frozen registry, falling back to normalised party-name similarity where the citation form differs.

Layer 2 : guarded abstention

Returns unclear where authoritative judgment text is unavailable, rather than inferring a holding from the citation alone. A deliberate design choice.

Layer 3 : passage similarity

TF-IDF cosine between the generated application and each retrieved passage. Thresholds 0.35 and 0.15, fixed before validation. Chosen as a transparent, reproducible baseline needing no external model access.

How it was tested

Predictions were compared with the adjudicated reference labels using exact accuracy, balanced accuracy, per-class precision, recall and F1, macro F1, confusion matrices and Cohen's Kappa [8].

Class frequencies are reported alongside accuracy because both the holding and grounding labels are imbalanced, and grounding performance is compared against an always-no baseline.

The V1 run is frozen as the primary result. A later V3 run supplied additional authority text for a post hoc sensitivity check.

Can the audit be automated?

Performance against the adjudicated reference labels for the 98 validation occurrences.

Layer	Accuracy	Kappa	Interpretation
Citation status	87.8%	0.80	Substantial agreement
Holding accuracy	4.1%	0.04	Abstained on all 94 assessable claims
Reasoning grounding	85.7%	0.24	Fair; matched by a majority-class baseline

The aggregate hides a weakness: the sample held seven retrieved-ID and citation-mismatch occurrences and the verifier identified none of them, so the citation-status result is driven by the two largest classes.

Holding accuracy: abstention, not failure

The verifier returned unclear for all 94 substantive occurrences because authoritative judgment text was unavailable. The 4.1% comes only from correctly classifying the four n-a rows.

Grounding: accuracy that proves nothing

84 of 98 labels were no, so always predicting no would also score 85.7%. The verifier found none of the 3 grounded and 3 of the 11 partially grounded occurrences.

**Citation provenance can be screened automatically.
Holding accuracy and grounding still require manual legal judgment.**

The risk lies beyond fabrication

No fabricated case was confirmed in this sample. A real citation still passes an existence check when the proposition attributed to it is wrong.

0

cases confirmed
as fabricated

50.0%

of assessable holdings
defective

85.7%

of reasoning
ungrounded

An audit based solely on whether a cited case exists would have given this system a clean bill of health. Yet a genuine authority lends apparent legitimacy to a proposition it does not establish, and that resists the superficial review a fabricated citation would not survive.

And it is not only a retrieval problem

Some misdescribed authorities were linked to retrieved material. Better retrieval may close the provenance gap, but it cannot ensure the generator describes a retrieved authority correctly.

Conclusions, limitations and future work

Conclusions

- No case was confirmed as fabricated, yet half of assessable holdings were defective and 85.7% of reasoning was ungrounded.
- Citation existence alone is an inadequate measure of legal-AI reliability.
- Automation works for provenance screening ($\kappa = 0.80$) but not for semantic accuracy.

Limitations

- 59 judgments, partly non-independent because of repeated generations.
- A single model and configuration; controlled experiments not conducted.
- R2 was AI-assisted, so agreement is not independent inter-human reliability. Grounding agreement was 40.8% ($\kappa = 0.054$).
- Holding and grounding labels are adjudicated, not objective ground truth.
- Grounding depends on each draft being linked to the correct retrieval event.

Future work

- Controlled experiments varying retrieval depth, prompt design and model.
- Supply authoritative full-text judgments so holding checks can be automated.
- A larger, multi-domain sample and an independent second human annotator.

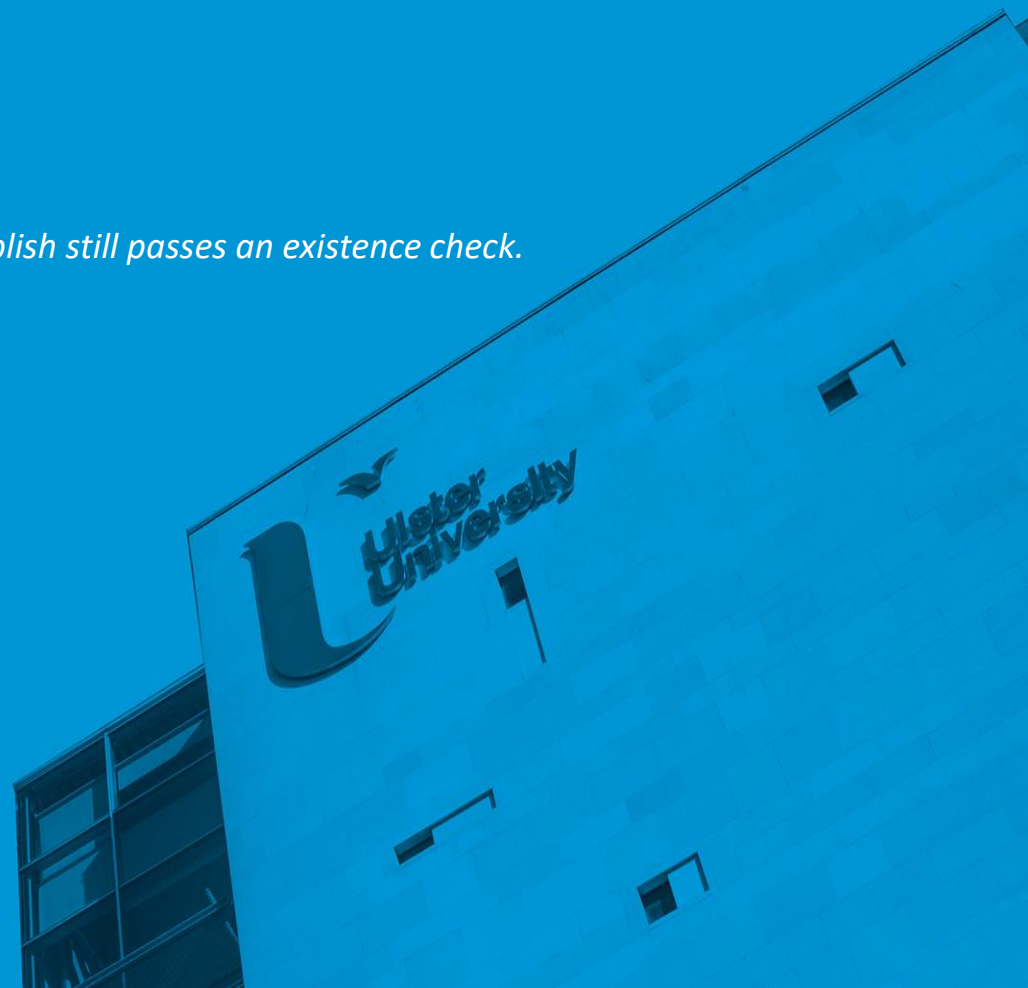
Safeguards for deployment: passage-level attribution for every proposition · abstention where authoritative text is unavailable · retrieval-provenance logging · human oversight aimed at the meaning of the authority, not merely its existence.

The central risk was not an invented case, but a real case cited for a proposition it does not establish.



Questions

A real case cited for a proposition it does not establish still passes an existence check.



References



- 1) V. Magesh et al., “Hallucination-free? Assessing the reliability of leading AI legal research tools,” arXiv:2405.20362, 2024.
- 2) M. Dahl, V. Magesh, M. Suzgun and D. E. Ho, “Large legal fictions: Profiling legal hallucinations in large language models,” J. Legal Analysis, vol. 16, no. 1, pp. 64–93, 2024.
- 3) Y. Gao et al., “Retrieval-augmented generation for large language models: A survey,” arXiv:2312.10997, 2024.
- 4) S. Es, J. James, L. Espinosa-Anke and S. Schockaert, “RAGAS: Automated evaluation of retrieval augmented generation,” Proc. EACL: System Demonstrations, 2024, pp. 150–158.
- 5) H. Rashkin et al., “Measuring attribution in natural language generation models,” Computational Linguistics, vol. 49, no. 4, pp. 777–806, 2023.
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- 8) J. Cohen, “A coefficient of agreement for nominal scales,” Educ. Psychol. Meas., vol. 20, no. 1, pp. 37–46, 1960.
- 9) A. Östling et al., “The Cambridge Law Corpus: A dataset for legal AI research,” Advances in Neural Information Processing Systems 36, 2023, pp. 41355–41385.

Backup: dataset composition

	Real	Synthetic	Total
Draft judgments	34	25	59
Citation occurrences	63	52	115
Case-law occurrences	61	49	110
Statutory / rule references	2	3	5
Validation judgments	29	21	50
Validation occurrences	55	43	98

Corpus: Qdrant knowledge base of 3,654 cases and 29,232 vector chunks; Neon/PostgreSQL registry of 2,541 structured case records. Underlying material includes records derived from the Cambridge Law Corpus [9].

Sampling: stratified by input type using SHA-256 ranking, seed 748, giving a deterministic and reproducible validation sample.

Backup: verifier detail and sensitivity check

Citation status. 86 of 98 correct; macro F1 = 0.71 across five validation classes. Strong on the two largest classes, not found in registry (F1 = 0.90) and in registry, not retrieved (F1 = 0.98). The sample contained seven retrieved-ID and citation-mismatch occurrences and the verifier identified none of them, so the aggregate is driven by the two dominant classes.

Holding accuracy. Returned unclear for all 94 substantive occurrences because authoritative judgment text was unavailable, an intentional safeguard against inferring holdings from citations alone.

Reasoning grounding. TF-IDF cosine similarity against retrieved passages, thresholds 0.35 (yes) and 0.15 (partial), fixed before validation. Predicted no for 92 occurrences, partial for 6, yes for none.

Post hoc V3 sensitivity check

Supplying the exported Qdrant authority text raised holding exact accuracy from 4.1% to 10.2%, while Kappa fell from 0.04 to roughly 0.002. Not a meaningful improvement, so V1 was retained as the primary result. The exported text is generated case summaries rather than judgment text, so V3 did not test automated holding verification against authoritative sources.

Annotation passes. Holding: 79 of 94 pairs agreed, 84.0%, $\kappa = 0.74$. Grounding: 40 of 98 agreed, 40.8%, $\kappa = 0.054$, with 54 of the 58 disagreements being R1 partial against R2 no. Descriptive only, R2 was AI-assisted.

Backup: recurring default authorities

Two authorities on the judicial duty to give reasons recurred across the corpus:

English v Emery Reimbold & Strick Ltd appeared in 22 of 59 judgments.

Flannery v Halifax Estate Agencies Ltd appeared in 25 of 59 judgments.

At least one of the two appeared in 30 of 59 judgments (50.8%).

Frequency did not bring accuracy

Among 21 labelled Flannery occurrences: 9 accurate, 8 partially accurate, 4 inaccurate. Even a heavily repeated authority was described correctly less than half the time.

The frequency suggests a possible default citation pattern rather than selection driven by the dispute in hand. Relevance to the underlying dispute was not annotated as a separate variable, so this is an observation for future work rather than a finding.